

MEDICAL CORE · VERIFIED

# rTMS Coil Navigation Software (NeuroPilot)

Turn optical tracking and 3D Slicer into a license-gated coil-navigation product that gives operators position and angle guidance they can rely on.

|                                                         |                                                                                 |
|---------------------------------------------------------|---------------------------------------------------------------------------------|
| <div>EVIDENCE STATE</div> <div>Verified · Ongoing</div> | <div>PERIOD</div> <div>2024.07 - present</div>                                  |
| <div>PORTFOLIO TIER</div> <div>Medical Core</div>       | <div>PRIMARY CAPABILITY</div> <div>Medical Navigation &amp; Visualization</div> |

Designed a 3D Slicer workflow from procedure setup through real-time coil navigation, a landmark-and-ICP registration engine, optical-tracker, tablet, and robot integration, and a license-gated product structure, delivered to the customer and maintained on an ongoing basis.

# Problem and system boundary

## 01 / NAVIGATION UI WORKFLOW

### Navigation UI workflow

Six Home-screen menus gate the sequence from loading DICOM through volume normalisation, skin and brain reconstruction, landmark registration, and target setup to the live session. The aiming screen carries

## PROBLEM

**Giving an operator coil position and angle guidance they can rely on meant tying medical images, a tracker, a tablet, and a robot, each in its**

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## PUBLIC BOUNDARY

**Although this is delivered and maintained medical navigation software, this case claims no clinical efficacy or quantitative accuracy, and excludes**

The brain-segmentation and landmark-detection models were supplied externally and integrated behind an adapter boundary; what I own is that boundary, the post-processing, the license gating, and the UI integration. Interfaces are agreed with the optical-tracker and robot side, the tablet-app developer, and one colleague.

## EVIDENCE STATE

Verified · Ongoing / NeuroPilot navigation module: tracker, patient-marker and coil-marker status, distance and tilt guidance to the target, and the aiming guide (project path hidden).

# Owned decisions and implementation

## MY ROLE

Led the 3D Slicer procedure-setup and navigation workflow, the registration engine combining landmarks and ICP with its coordinate-transform chain, the optical-tracker SDK migration with the tablet protocol and robot integration, and the product structure covering license verification and optional-feature gating.

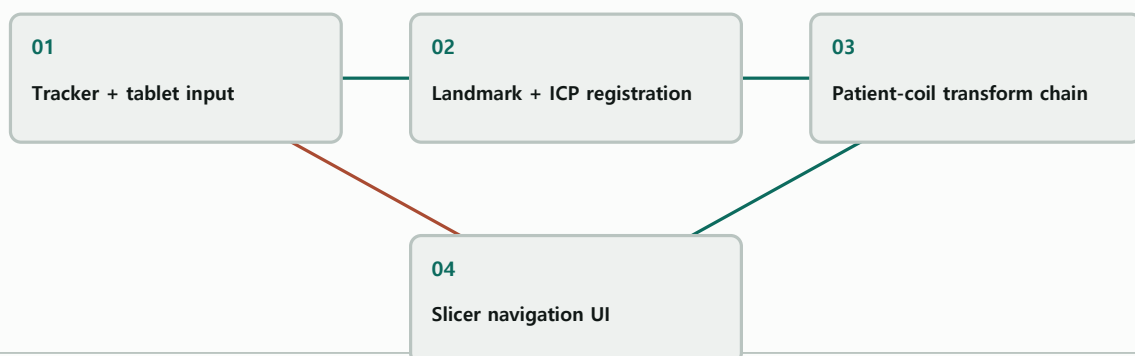
## 02 / COORDINATE AND REGISTRATION ENGINE

### Coordinate and registration engine

The transform chain from image coordinates through patient, tracker, and coil frames stays explicit in both

## SYSTEM RELATIONSHIP

### Coordinate loop from input to aiming guidance



Explanatory system relationship diagram; it is not photographic or experimental evidence.

## IMPLEMENTATION TECHNOLOGIES

3D Slicer / VTK / PyQt5 / Python / PyTorch / Optical tracking / TCP/IP binary protocol / License gating

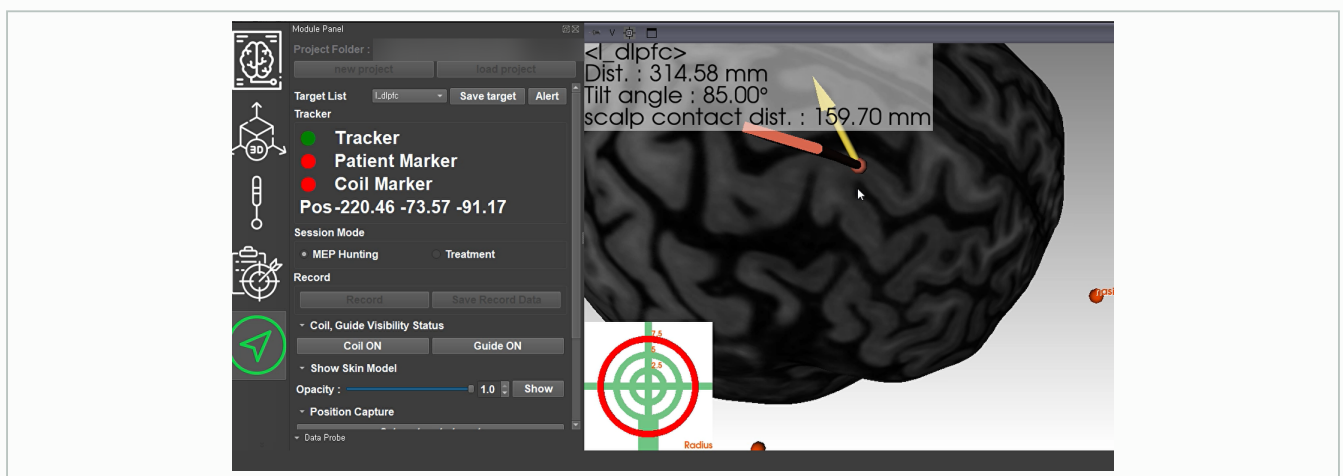
# Verification and evidence ledger

## 03 / DEVICE AND SYSTEM INTEGRATION

### Device and system integration

Migrated the optical-tracker SDK across a major version and talk to the tablet over an in-house TCP binary protocol. Tracker ownership is concentrated in one component so screens no longer contend for its state,

#### IMAGE / PUBLICLY INSPECTABLE



#### REGISTERED PUBLIC EVIDENCE

##### IMAGE / PUBLICLY INSPECTABLE

### rtms-navigation-lead-01

Approved derivative; caption in portfolio data.

##### IMAGE / PUBLICLY INSPECTABLE

### rtms-navigation-gallery-01

Approved derivative; caption in portfolio data.

#### VERIFICATION AND EVIDENCE LEDGER

The evidence is the license-gated deployment build, a regression-test suite grown issue by issue, and coordinate-transform and registration logs.

# Role, team result, and limits

## 04 / CLINICAL AND PRODUCT BOUNDARY

### Clinical and product boundary

Although the product is licensed, delivered, and maintained, no clinical efficacy or quantitative accuracy is

## MY ROLE

### Led the workflow, registration engine, device integration, and product structure.

Led the 3D Slicer procedure-setup and navigation workflow, the registration engine combining landmarks and ICP with its coordinate-transform chain, the optical-tracker SDK migration with the tablet protocol and robot integration, and the product structure covering license verification and optional-feature gating.

## TEAM RESULT

### The core registration, coordinate-transform, and navigation-UI structure was designed and built individually, with one colleague contributing to

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## LIMITATIONS

**Although this is delivered and maintained medical navigation software, this case claims no clinical efficacy or quantitative accuracy, and excludes regulatory status, patient data, and hospital-identifying information.**

NeuroPilot navigation module: tracker, patient-marker and coil-marker status, distance and tilt guidance to the target, and the aiming guide (project path hidden).

## COLLABORATION BOUNDARY

**The brain-segmentation and landmark-detection models were supplied externally and integrated behind an adapter boundary; what I own is that boundary, the post-processing, the license gating, and the UI integration. Interfaces are agreed with**

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# Recap and joint-development invitation

**Turn optical tracking and 3D Slicer into a license-gated coil-navigation product that gives operators position and angle guidance they can rely on.**

## PRIMARY CAPABILITY

Medical Navigation & Visualization / 3D Geometry & Registration

## REGISTERED PUBLIC EVIDENCE

No external public link is currently registered.

## WHAT TO BRING

**Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.**

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